

What are the frequently used annotations in spring boot?

Great question, Srikanth! Let's break this down tutor-style - with clarity, examples, and a memory hook so you won't forget.

What Are Spring Boot Annotations?

Annotations in Spring Boot are like **labels** or **tags** that tell the framework how to behave. They reduce boilerplate code and make your app smarter with less effort.

Easy-to-Remember Core Annotations

Here's a list of frequently used annotations, grouped by purpose, with simple explanations and examples:

Application Setup

Annotation	Purpose	Example
@SpringBootApplication	Main entry point; combines 3 annotations: @Configuration, @EnableAutoConfiguration, @ComponentScan	@SpringBootApplication public class App {}

Memory Tip: Think of this as the “Start Button” of your app.

Dependency Injection

Annotation	Purpose	Example
@Autowired	Injects dependencies automatically	@Autowired private MyService service;
@Qualifier	Specifies which bean to inject when multiple exist	@Autowired @Qualifier("myBean")

Memory Tip: “Auto” + “Wired” = Automatically connects your beans.

Bean & Component Declaration

Annotation	Purpose	Example
@Component	Generic bean for Spring container	@Component public class MyBean {}
@Service	Business logic layer	@Service public class MyService {}
@Repository	Data access layer	@Repository public class MyRepo {}
@Configuration	Declares a configuration class	@Configuration public class AppConfig {}

Annotation	Purpose	Example
@Bean	Declares a bean inside config class	@Bean public MyService service() {}

Memory Tip: Think of these as **roles** in your app—like actors in a play.

Web Layer

Annotation	Purpose	Example
@Controller	Handles web requests (MVC)	@Controller public class WebController {}
@RestController	Combines @Controller + @ResponseBody for REST APIs	@RestController public class ApiController {}
@RequestMapping	Maps HTTP requests to methods	@RequestMapping("/home")
@GetMapping, @PostMapping	Shortcut for HTTP methods	@GetMapping("/hello")

Memory Tip: REST = @RestController, GET = @GetMapping, POST = @PostMapping.

Bonus: Configuration & Lifecycle

Annotation	Purpose
@Value	Injects values from properties file
@EnableAutoConfiguration	Enables auto config based on classpath
@PostConstruct	Runs after bean initialization
@PreDestroy	Runs before bean destruction

Mnemonic to Remember

SACRED WARP — a fun acronym to remember key annotations:

- SpringBootApplication
- Autowired
- Component
- RestController
- EnableAutoConfiguration
- Dependency Injection (@Qualifier)
- Web Mapping (@RequestMapping, @GetMapping)
- Actor Roles (@Service, @Repository)
- Request Lifecycle (@PostConstruct)
- Properties (@Value)

